

Test Edge Cases and Fairness

Project: Helpful Classroom Sorter

1. Test Edge Cases and Fairness

Teach the learning cycle

Primary target: Choose tests just below, at and above a decision boundary

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Retrieval answer guidance

Test Normal Cases

Without opening the old lesson, explain one key idea from Test Normal Cases. Give an example and a mistake that the idea helps you avoid.

Look for: Plan routine cases with expected results derived from the requirements

Reference example, not the only acceptable wording: R1 B1 because time fits and it is quiet; early match. R2 B2 because B1 is too long, B2 fits and quiet is unrestricted; later match. R3 no match because every time exceeds 6; exhaustion. R4 no match because B2 fails quiet while other rows exceed 9; both conditions. Record P1 version 1, B version 1 and the manual baseline. A full pass needs the required output and a correct explanation.

Design the Decision Process

Without opening the old lesson, explain one key idea from Design the Decision Process. Give an example and a mistake that the idea helps you avoid.

Look for: Design a decision process that covers valid, invalid and no-match requests

Reference example, not the only acceptable wording: Clarification follows invalid or missing input. A suggestion follows the first candidate passing both compatibility conditions. No match follows a valid request after all candidates fail. A single failed candidate does not settle whether later candidates fit, so the process must advance.

Humans Set Boundaries

Without opening the old lesson, explain one key idea from Humans Set Boundaries. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish a system's capability, the correctness of its output and permission to take an action

Reference example, not the only acceptable wording: The button establishes a capability. The fact check establishes factual support for A at that time. The approval permits the specified classroom-board action within its limits. Public posting is outside that permission even though the button exists and A is factually checked.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Before testing, define outcomes for clean paper, metal foil, mixed packaging and an unreadable card. The team later changes mixed packaging's expected result to match its sorter. Explain the evaluation problem.

Concept to explain

Unusual and underrepresented cases reveal hidden assumptions.

Term · Meaning

Term: edge case

Meaning: unusual input near a system limit

Term: underrepresented

Meaning: rare or absent in design evidence

Term: subgroup test

Meaning: check for one relevant condition or user group

Term: hidden assumption

Meaning: unstated belief revealed by failure

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Boundary testing returns to inclusive comparisons and validation using the capstone context. S version 0 is a deliberately faulty comparison example, not the correct P1 implementation. It can pass the earlier routine set while rejecting exact duration fits. Catalogue variations test later successful paths that the original data cannot reach. Preserve the distinction between changing data, changing requirements and changing implementation; the variations here change only declared catalogue values.

Readiness answer

They examine only the selected inputs and conditions. A mistake may affect equality, invalid input, another catalogue or a path that none of those cases reaches.

Saved P1 instructions and catalogue B version 1, plus separate labelled copies for versions 2 and 3

Test records with purpose, versions, input, expected result, observed result, judgement and trace

Paper cards for the independent equipment selector and a folder retaining every attempted run

End goal for the pictorial case

By the end, I can design boundary and coverage tests instead of assuming routine success is enough.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

2. Explain the mechanism

Explain the mechanism

Follow the information instead of imagining magic inside the machine.

INPUT planned edge and subgroup cases

PROCESS run same version and compare error patterns

OUTPUT documented gaps, harms, and repair priorities

CORE IDEA Unusual and underrepresented cases reveal hidden assumptions.

UNDER THE HOOD Edge testing explores boundaries, rare combinations, missing values, adversarial ambiguity, and subgroup conditions. Fairness analysis asks whether error burdens differ and whether fallback paths remain usable.

Draw a fresh example with all three stages:

Every step connects a claim to observable evidence.

CASE A worn notebook, multilingual label, partly hidden item, and object fitting two bins reveal assumptions normal tests miss.

STEP 1 — OBSERVE planned edge and subgroup cases

STEP 2 — TRACE run same version and compare error patterns

STEP 3 — CONCLUDE documented gaps, harms, and repair priorities

MODEL REASONING Use the same frozen prototype, log outcomes by condition, and compare error types. Prioritize failures with high burden or no usable fallback, then design a targeted repair without stereotyping users.

DO NOT OVERCLAIM Edge cases are not excuses to blame users for being unusual; they are evidence about design limits.

What new evidence could change this answer?

Concrete teaching case

A sorter handles clean paper but fails on crumpled paper and mixed packaging.

Design revealing tests.

Exact explanation to model

Use separately labelled clean, crumpled and mixed cases with expected outputs or review actions declared first. Compare conditions.

Explain the added worked example

Routine examples can miss a mistake at an exact limit. This week you will choose revealing cases around those limits and vary the paper catalogue to reach later successful outputs. Write expected results from the requirements before operating the prototype. Keep its saved instructions and each test record so a failure remains useful evidence.

Catalogue B version 1 is unchanged: B1 Bookmark drawing, 10 minutes, quiet Yes; B2 Story circle, 8 minutes, quiet No; B3 Pattern puzzle, 12 minutes, quiet Yes; B4 Tower challenge, 15 minutes, quiet No. Search in that order. These fictional paper values assume materials are available. Inputs are whole minutes from 0 to 30 inclusive and Quiet required Yes or No. No allows quiet or nonquiet activities.

Time fits when required minutes are no more than available minutes, including equality. The quiet condition must also hold. Select the first fitting row with a correct reason, or report no match after every row fails. Invalid or missing input requires clarification before searching. P1 validates, resets to the first row, checks both conditions, advances after failure and stops at the first fit or catalogue exhaustion. No output books or publishes anything.

For quiet requests, 10 minutes is the first time a catalogue activity fits. Test 9, 10 and 11 with Yes so only the time changes. A faulty comparison S version 0 uses strictly less than instead of no more than for time, while keeping the other P1 rules. It rejects B1 at exactly 10. With quiet required, every later row also fails, so it incorrectly reports no match.

S can pass all four routine requests from Week 33: 11 Yes, 9 No, 6 No and 9 Yes. None has a fitting activity whose duration equals the request. The 10 Yes case reveals a gap those passes did not close. Another useful boundary is 8 minutes with No: equality should allow B2, while 7 No has no match. Keep both quiet values in your wider plan because No does not mean nonquiet only.

Key terms

Term · Meaning

Term: Boundary

Meaning: A value where an input changes category or a condition changes truth

Term: Inclusive

Meaning: Including the stated endpoint or equal value

Term: Invalid input

Meaning: A value that does not meet the agreed input rules

Term: Test variation

Meaning: A documented change in the conditions used for a test

Term: Coverage gap

Meaning: A relevant condition or behaviour the current tests have not examined

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

0 minutes Yes · No match · Zero is valid; every activity needs more time

0 minutes Yes: 30 minutes Yes

No match: B1 with a matching reason

Zero is valid; every activity needs more time: 30 is valid; B1 is the first fit

0 minutes Yes: 31 minutes Yes

No match: Clarify minutes; no search

Zero is valid; every activity needs more time: 31 is outside the accepted range

0 minutes Yes: Blank minutes with Yes

No match: Clarify minutes; no search

Zero is valid; every activity needs more time: The time field is missing

0 minutes Yes: 10.5 minutes Yes

No match: Clarify minutes; no search

Zero is valid; every activity needs more time: The accepted time must be a whole number

0 minutes Yes: 10 minutes with blank quiet

No match: Clarify quiet; no search

Zero is valid; every activity needs more time: Choose Yes or No before searching

The accepted range and activity durations are different kinds of boundary. Zero belongs to the valid range even though no activity fits. Thirty also belongs to it. A negative time, such as minus one minute, is below zero and invalid. Do not change a blank to zero, round 10.5 to 10 or turn a blank quiet choice into No. Ask for the field to be corrected and start a new validated run.

A test record should say whether a search is expected to begin. Clarification should name the field needing correction; it should not claim the catalogue has no suitable activity. Test an unrecognised quiet value such as Maybe as well as a missing value. These check different forms of unusable input.

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Test Edge Cases and Fairness

A fictional craft-book chooser accepts requests for whole available minutes from 0 to 20.

The facts we will use

Booklet A takes 6 minutes; B takes 12. Return first duration at most available time.

Check 5,6,7; also 0,20,21 and blank. Same catalogue applies to every user.

Test Edge Cases and Fairness

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Near a boundary

At 5: none. At 6 and 7: A.



2. Valid extremes

0: none. 20: A, because first-fit order remains fixed.



3. Invalid inputs

21 and blank require clarification, not ordinary search.



4. Coverage record

Record each user condition tested; do not invent unrun outcomes.

CHANGE ONE THING

Faulty code uses duration strictly below available time.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Near a boundary

What happens and why: At 5: none. At 6 and 7: A.

Step: 2. Valid extremes

What happens and why: 0: none. 20: A, because first-fit order remains fixed.

Step: 3. Invalid inputs

What happens and why: 21 and blank require clarification, not ordinary search.

Step: 4. Coverage record

What happens and why: Record each user condition tested; do not invent unrun outcomes.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Adjacent values below, at and above 6 expose an accidental strict inequality. Zero is valid even though no activity fits. The top valid input does not require selecting the longest activity: catalogue order still controls selection. Coverage also asks whose conditions were tested, which cannot be inferred from these numbers alone.

What this example does not establish

Equal written rules do not prove equally good outcomes for every group; actual condition-specific evidence is still needed.

Change one thing in the pictured case

Change: Faulty code uses duration strictly below available time.

Prediction: Request 6 now wrongly returns none.

Reason: A's duration equals the available time and should be eligible under at most.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Demonstrate and guide practice

Board sequence

Write the given facts: A sorter handles clean paper but fails on crumpled paper and mixed packaging.

Write the question to explain: Design revealing tests.

Reveal after predictions: Use separately labelled clean, crumpled and mixed cases with expected outputs

or review actions declared first. Compare conditions.

Keep for an independent attempt: A sorter works on clear photos. Test a blurred pencil, a dim book and an unfamiliar lunch box. What makes these useful tests?

Classroom dialogue

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks evidence; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

For B version 3 and 15 No, B1 20, B2 18 and B3 17 fail time. B4 15 fits exactly and quiet is unrestricted, so select B4 and stop. For 15 Yes, B4 fails quiet and all earlier rows fail time, so no match after all four. Record B version 3, the saved instruction version, exact request, expected and observed output with reasons, and trace. Exit: inspect strictly less than versus no more than; the specification includes equality, so expected 10 Yes remains B1. Another equality case is 8 No, expected B2. Invalid 31 Yes requires minutes clarification before searching.

Case-specific teaching sequence

Stage · Teacher action

Stage: Session 1 start to 7 minutes

Teacher action: Retrieve why routine-case success can leave coverage gaps.

Stage: 7 to 20 minutes

Teacher action: Use the diagram and W1 to compare below, at and above ten minutes.

Stage: 20 to 33 minutes

Teacher action: Complete W2; distinguish accepted range endpoints from activity duration limits.

Stage: 33 to 45 minutes

Teacher action: Prepare B version 2 and begin W3 expected results and partner traces.

Stage: Session 2 start to 8 minutes

Teacher action: Finish W3 and verify both methods used the same catalogue copy.

Stage: 8 to 23 minutes

Teacher action: Trace guided version 3 requests and discuss purpose and version labels.

Stage: 23 to 38 minutes

Teacher action: Collect W4 independently before support; inspect both equality cases and validation.

Stage: 38 to 45 minutes

Teacher action: Complete W5 and the exit; save the expanded development set for repair checks.

Reduce help deliberately

Completion task: Classify these requests before searching: 0 Yes, 30 Yes, 31 Yes, blank minutes with Yes, 10.5 Yes, and 10 Maybe. Give each expected output or clarification and explain whether any candidate should be checked. Why must zero not be treated like a blank?

Teacher check: 0 Yes is valid and checks all rows before no match. 30 Yes is valid and selects B1 at the first row. 31, blank minutes and 10.5 require minutes clarification; 10 Maybe requires a Yes or No quiet choice. Invalid cases check zero candidates. Zero is a supplied whole number within range; blank is missing information.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Testing boundaries and varied conditions. Proposed explanation: “At the limit means outside the rule”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: An inclusive rule accepts its endpoints and equality.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more

demanding variation.

Pictorial practice question

Which single boundary run most directly detects the strict-comparison bug, and why?

Reasoned teacher answer

6: intended A, faulty none. Inputs 5 and 7 alone would miss this exact equality defect.

Changed-case reasoning: Request 6 now wrongly returns none. A's duration equals the available time and should be eligible under at most.

Collect each learner's explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the weekly evidence lab

Run the weekly evidence lab

Predict first, test honestly, and explain what the evidence means.

State your prediction before the result is known.

Complete the task: Add missing, ambiguous, and subgroup test cards.

Keep a normal case and an edge case clearly separated.

Record what happened—even when it does not match your prediction.

Explain the result, one limitation, and the next useful test.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing. Keep the expected result separate from an observed mistake.

Record · What to capture

Record: Input or evidence

What to capture: Exact data, instruction or claim

Record: Expected result

What to capture: Answer or safe fallback declared before the run

Record: Observed result

What to capture: Actual result, including errors

Record: Explanation

What to capture: Rule, calculation or evidence explaining the outcome

Record: Next test

What to capture: One justified change and a new prediction

Diagnose and reteach

CLAIM TO REPAIR Repeating only easy inputs tests the system's limits.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation. Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

At the limit means outside the rule: An inclusive rule accepts its endpoints and equality.

No match and invalid mean the same thing: No match follows valid input; invalid fields need clarification before search.

A catalogue change needs the same expected answer: Recompute expectations from the declared new data and unchanged requirements.

More routine repetitions close every gap: Choose cases that exercise a missing condition or path.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Assess and explain the answers

Worked case answer

Use separately labelled clean, crumpled and mixed cases with expected outputs or review actions declared first. Compare conditions.

Independent question

A sorter works on clear photos. Test a blurred pencil, a dim book and an unfamiliar lunch box. What makes these useful tests?

Expected independent answer

They probe unclear input, changed lighting and an unknown category. Record expected behaviour before each test, including review when evidence is insufficient.

What counts as evidence

Criterion · 0 points · 1 point · 2 points

Criterion: Result

0 points: Incorrect or absent

1 point: Partly correct

2 points: Correct with necessary conditions

Criterion: Reasoning

0 points: No supporting trace

1 point: Some steps

2 points: Clear mechanism, calculation or source support

Criterion: Limit

0 points: Overclaims certainty

1 point: Vague limitation

2 points: Relevant uncertainty or missing fact

Criterion: Transfer

0 points: Copies without adapting

1 point: Adapts with prompts

2 points: Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold. Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the evidence that challenges this claim: “Repeating only easy inputs tests the system's limits.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

A reasonable policy is P, M, review, review. Changing expected labels just to match predictions makes the test circular. If the policy itself was wrong, document a justified correction and reevaluate all affected

cases, preserving the original record.

Explanations for the additional workbook examples

W1 Plan the equality tests

For catalogue B version 1, write expected outputs and reasons for 9 Yes, 10 Yes and 11 Yes. Explain why keeping quiet and catalogue fixed helps examine the time boundary. Predict what the faulty strictly-less-than S comparison does for 10 Yes.

9 Yes has no match: B1 and B3 are too long and B2 is not quiet; B4 also fails. 10 Yes and 11 Yes select B1, which is quiet and fits at equality or with time left. Fixing the other conditions isolates the time comparison. S rejects equality at B1 and cannot find a later fit for 10 Yes, so incorrectly returns no match.

W2 Check the validation boundary

Classify these requests before searching: 0 Yes, 30 Yes, 31 Yes, blank minutes with Yes, 10.5 Yes, and 10 Maybe. Give each expected output or clarification and explain whether any candidate should be checked. Why must zero not be treated like a blank?

0 Yes is valid and checks all rows before no match. 30 Yes is valid and selects B1 at the first row. 31, blank minutes and 10.5 require minutes clarification; 10 Maybe requires a Yes or No quiet choice. Invalid cases check zero candidates. Zero is a supplied whole number within range; blank is missing information.

W3 Run a labelled catalogue variation

Copy B version 1 and change only B1's duration to 20, labelling it B version 2. Write the expected result for 12 Yes, then run your saved P1 and the manual baseline on this copy. Record the actual traces separately. Explain why an answer checked against B version 1 would not be a fair judgement of this run.

Expected B3: B1 takes too long, B2 fails quiet, B3 fits 12 exactly and is quiet. Correct P1 and baseline agree, but retain actual disagreements for investigation. Under version 1 the expected first fit is B1, so using that expectation with changed data would mix conditions. Save catalogue and instruction versions with both traces.

W4 Find an independent capacity boundary

A new equipment selector lists E1 first, capacity 4; E2 second, capacity 8. Valid group size is a whole number from 1 to 10. Choose the first item whose capacity is at least the group size, or no match; invalid input needs clarification. A faulty version uses capacity strictly greater than group size. Independently give correct and faulty outputs for groups 3, 4 and 5, then for group 8. Explain the defect and give the correct response to group 0. No booking occurs.

Correct outputs for 3, 4, 5 and 8 are E1, E1, E2 and E2. Faulty outputs are E1, E2, E2 and no match. Equality is wrongly excluded at capacities 4 and 8. Use capacity at least group size, including equality. Group 0 is outside the accepted 1 to 10 range and needs clarification before any search. This range differs from the activity time range.

W5 Audit what changed in the test

A team runs 15 No on B version 3, gets B4, but records “B version 1” and calls B4 correct. Explain the documentation error and the expected result under each catalogue. State what must stay the same in a fair prototype-versus-baseline comparison and why a test not yet run cannot be counted as a pass.

For version 3, B1 20, B2 18 and B3 17 are too long; B4 15 fits and quiet is unrestricted, so B4 is correct. Under version 1, earlier B1 fits and must be selected. Correct the catalogue label while preserving what was actually used and observed. Both methods need the same request, catalogue and pass criterion. An unrun case has no observed evidence, so cannot be counted as passing.

Assess independent transfer

Selected case: A new equipment selector lists E1 first, capacity 4; E2 second, capacity 8. Valid group size is a whole number from 1 to 10. Choose the first item whose capacity is at least the group size, or no match; invalid input needs clarification. A faulty version uses capacity strictly greater than group size. Independently give correct and faulty outputs for groups 3, 4 and 5, then for group 8. Explain the defect and give the correct response to group 0. No booking occurs.

Selected case answer

Correct outputs for 3, 4, 5 and 8 are E1, E1, E2 and E2. Faulty outputs are E1, E2, E2 and no match. Equality is wrongly excluded at capacities 4 and 8. Use capacity at least group size, including equality. Group 0 is outside the accepted 1 to 10 range and needs clarification before any search. This range differs from the activity time range.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect and extend learning

Connect and extend learning

Helpful Classroom Sorter

Expand the sorter test suite with edge, missing-data, and fairness cases plus expected safe behavior.

Keep the decision, input, result, explanation and remaining limitation in the project evidence record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use physical cards, spoken explanations and a three-stage drawing. Read aloud without giving the answer. Stretch: compare two mechanisms with the same visible output and explain what evidence distinguishes them.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more evidence or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 35: Improve One Cause. Retrieve today's evidence and identify what carries forward and what is new.

Support and fresh reassessment

Draw a number line around each boundary and read at least or no more than aloud. Keep validation and candidate checks on separate cards. Provide version-label headings without filling expected results.

Extension: contrast a catalogue reorder with a duration change and explain how the first-match requirement remains unchanged in both.

W4 should distinguish correct E1, E1, E2, E2 from faulty E1, E2, E2, no match for 3, 4, 5, 8. Require an inclusive-capacity explanation and clarification for zero, not the activity task's valid-zero rule. After support, use a new catalogue F1 capacity 5 then F2 capacity 9 with the same 1 to 10 input range. For groups 5 and 9, correct F1 and F2; strict comparison gives F2 and no match. Require an independent explanation of both changes.

Week 35 repairs a documented defect, reruns revealing cases and checks earlier successful cases for

regressions before fixing a version for final evaluation.

Plan the delayed return

Suggested return points in this level: ai-1/35. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Add sorter boundary cases, missing fields and relevant user conditions to the test plan before making

fairness claims.

7. Prepare, teach and assess

Teaching target: Choose tests just below, at and above a decision boundary

Prepare the labelled worked-case picture, the matching workbook, fictional input cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

Session: Session 1 · Understand · 45 min

Plan: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Plan: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Plan: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Classify these requests before searching: 0 Yes, 30 Yes, 31 Yes, blank minutes with Yes, 10.5 Yes, and 10 Maybe. Give each expected output or clarification and explain whether any candidate should be checked. Why must zero not be treated like a blank?

Reasoned practice answer: 0 Yes is valid and checks all rows before no match. 30 Yes is valid and selects B1 at the first row. 31, blank minutes and 10.5 require minutes clarification; 10 Maybe requires a Yes or No quiet choice. Invalid cases check zero candidates. Zero is a supplied whole number within range; blank is missing information.

Check every learner before discussion: Use the worked case for Testing boundaries and varied conditions. Proposed explanation: “At the limit means outside the rule”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: An inclusive rule accepts its endpoints and equality.

Support: read the exact case aloud, enlarge the labelled picture, supply input cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Faulty code uses duration strictly below available time. Require a prediction and a reason before comparing. Expected result: Request 6 now wrongly returns none. Reason: A's duration equals the available time and should be eligible under at most.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion · Not yet · With support · Independent

Criterion: Explanation

Not yet: Names an answer without a deciding fact

With support: Uses a hint to link fact and decision

Independent: Explains the deciding fact in own words

Criterion: Trace or test

Not yet: Cannot connect input and result

With support: Completes steps with prompts

Independent: Traces or tests a changed case accurately

Criterion: Limits and revision

Not yet: Claims more than evidence supports

With support: Names a limit after a prompt

Independent: States a defensible limit and useful next test

Project checkpoint: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with evidence.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and evidence.